

Multiple S isotopes, zircon Hf isotopes, whole-rock Sr-Nd isotopes, and spatial variations of PGE tenors in the Jinchuan Ni-Cu-PGE deposit, NW China

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Received: 14 April 2015 / Accepted: 16 November 2015 / Published online: 27 November 2015
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Abstract Previous geochemical data for the Jinchuan Ni-Cu-(platinum-group elements, PGE) deposit, the single largest magmatic sulfide deposit in the world, are derived primarily from the upper parts of the deposit. This paper reports new PGE and S-Hf-Sr-Nd isotope data for the lower parts of the deposit that have become accessible for sampling by ongoing underground mining activity. New PGE data from this study, together with previous results, indicate that PGE tenors in the bulk sulfide ores of the deposit increase eastward, except for two fault-offset ore zones which occur together within the western part of the deposit. Generally, these two ore zones show depletions in IPGE (Ir, Ru, Rh) but not in PPGE (Pt, Pd) and Cu, and more fractionated olivine and Cr-spinel compositions than the rest of the deposit. These differences can be explained by a more evolved parental magma for the IPGE-depleted ore zones. The eastward increase of PGE tenors in the rest of the deposit can be explained by upgrading of preexisting sulfide liquid in a subhorizontal conduit by a new surge of magma moving through the conduit from west to east, which took place before the formation of the IPGE-depleted ore zones. The Jinchuan ultramafic rocks are

characterized by elevated initial $^{87}\text{Sr}/^{86}\text{Sr}$ ratios from 0.7077 to 0.7093, negative ε_{Nd} values from -9.2 to -10.5 , and zircon ε_{Hf} values from -4 to -7 . These data indicate up to 20 % of crustal contamination in the Jinchuan magma. Four of nine multiple sulfur isotope analyses for the Jinchuan deposit show anomalous $\Delta^{33}\text{S}$ values varying from 0.12 to 2.67‰. These results, together with elevated $\delta^{34}\text{S}$ values ($>2\%$) for some of the samples analyzed previously by other researchers, indicate the involvement of external sulfur from Archean and Proterozoic sedimentary rocks. Modeling results based on our olivine data and magma compositions estimated previously by other researchers indicate that fractional crystallization did not play a major role in triggering sulfide saturation in the Jinchuan magma and that the parental magma experienced previous sulfide segregation during ascent or in staging chambers.

Keywords PGE · Hf-Sr-Nd isotopes · Multiple sulfur isotopes · Magmatic sulfides · Ultramafic rocks · Jinchuan deposit · China

Introduction

The Jinchuan Ni-Cu-(platinum-group elements, PGE) sulfide deposit is one of the three largest magmatic sulfide deposit camps and the largest single magmatic sulfide deposit in the world (Naldrett 2011). It contains more than 500 million metric tons of ores with 1.1 wt% Ni and 0.7 wt% Cu (Tang et al. 2009). Recently, many studies on the deposit and its host intrusion have been carried out (e.g., Chai and Naldrett 1992a, b; Chen et al. 2013; de Waal et al. 2004; Li et al. 2004; Lehmann et al. 2007; Song et al. 2006, 2009, 2012; Su et al. 2008; Tang 1993; Tonnelier 2010). These studies have improved our understanding of the genetic relationship between magma evolution and ore genesis. However, the

Editorial handling: M. Fiorentini

Electronic supplementary material The online version of this article (doi:10.1007/s00126-015-0626-8) contains supplementary material, which is available to authorized users.

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previous studies all relied on samples from shallow depths. As a result, the spatial variations of olivine compositions and PGE tenors in the entire deposit, including the deeper parts of the deposit, are not available. In this paper, we use new data for the lower parts of the deposit that are now accessible for sampling due to ongoing underground mining operations, together with the existing data for its upper parts, to show a more complete picture of the deposit in olivine compositional variation as well PGE tenor variation. The fundamental controls on such variations are then discussed. Some of the new samples have also been analyzed for Sr-Nd isotopes and multiple sulfur isotopes. Zircon crystals separated from a large sample from the host intrusion have been analyzed for Hf isotopes. The integrated olivine and isotope data are used to evaluate the role of crustal contamination versus fractional crystallization in triggering sulfide saturation in the Jinchuan magma.

Geological background

The geology of the Jinchuan magmatic Ni-Cu sulfide deposit was described in detail by Tang (1990, 1993), Tang and Li (1995), Chai and Naldrett (1992a, b). The host intrusion is located in southern Alxa, the westernmost part of the North China Craton (Fig. 1a). Available data show that the 1.9–2.7 Ga amphibolites and granitic gneisses are the oldest rocks in this region (Geng et al. 2006; Gong et al. 2012). The main

landmark in southern Alxa is a NE-trending mountain range called Longshou Mt, which is commonly referred to as the Longshoushan terrane in the geological literature (Fig. 1b). The Longshoushan terrane is mainly composed of Proterozoic metamorphic rocks (migmatites, gneisses, and marbles), Paleozoic metasedimentary rocks (slates and sandstones), and Paleozoic granitoids. The granitoids are not restricted to the Longshoushan terrane but also occur in the adjacent North Qilian Orogenic Belt to the south (Fig. 1b). The collision between North Qilian and southern Alxa took place in the Paleozoic, as indicated by ~450 Ma eclogites in the suture (Song et al. 2013).

Many mafic-ultramafic intrusions are present in the Longshoushan terrane (Fig. 1b). Available zircon U-Pb isotope ages indicate two episodes of mafic magmatism in the region. The older one took place at ~830 Ma, as represented by the Jinchuan ultramafic intrusion (Li et al. 2005; Zhang et al. 2010). The younger one occurred at ~420 Ma, as represented by the Xijing mafic intrusion (Duan et al. 2015a) and mafic dykes in the Jinchuan area (Duan et al. 2015b). The ages of other mafic-ultramafic intrusions in the Longshoushan terrane are still not known.

The Jinchuan ultramafic intrusion intruded Proterozoic marbles and gneisses. It is ~6000 m long and 300 m wide, with downward extension >1000 m from the surface (Fig. 2). The current form of the intrusion looks like a SE-trending dyke, subparallel to the regional structural lineament (Figs. 1b and 2a), with steep dips toward the NW (Fig. 2c).

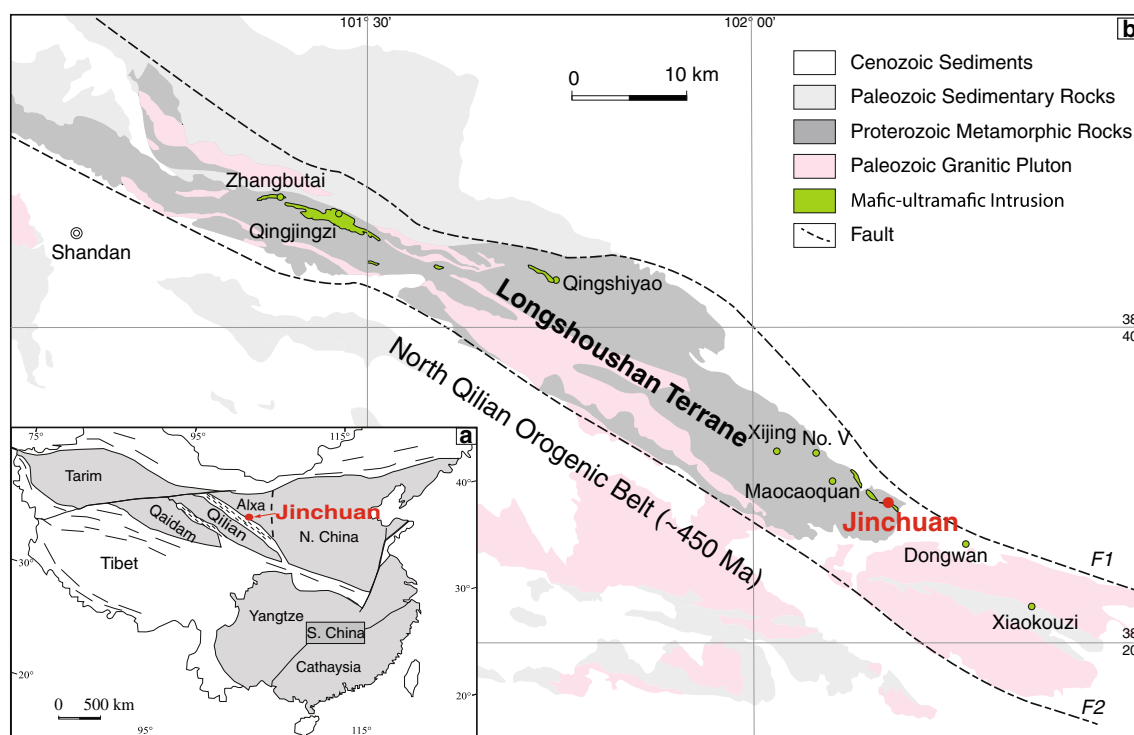
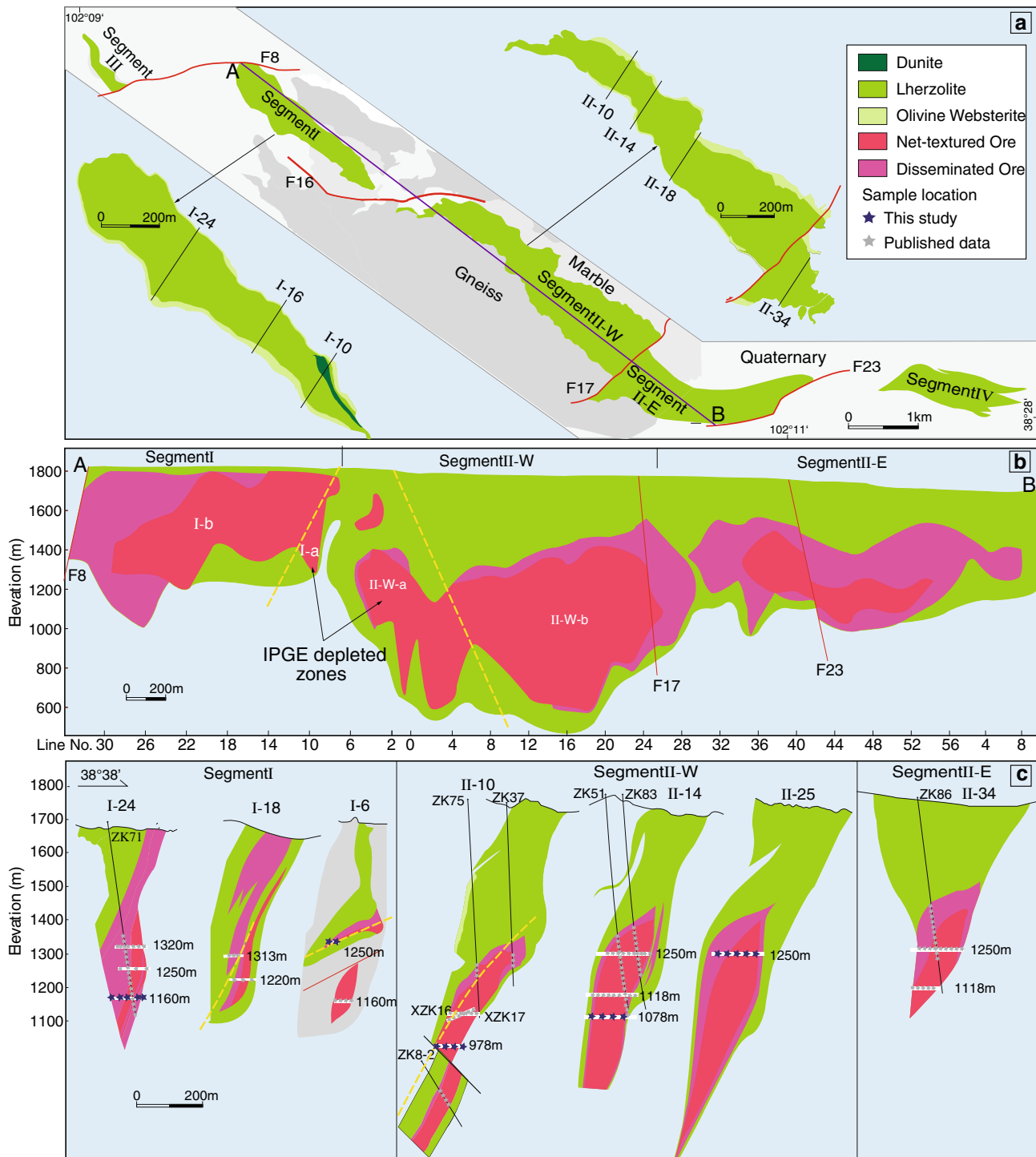


Fig. 1 The location of the Jinchuan Ni-Cu deposit in China (a) and simplified geologic map of the Longshoushan terrane (b)



websterite occurs locally in the margins of the intrusion (Fig. 2a); a small dunite zone is found only in segment I close to its NE edge (Fig. 2a). In drill cores, however, dunite is more common, especially in the sulfide-mineralized zones (de Waal et al. 2004).

Overall, lherzolite is the most abundant rock type in the Jinchuan intrusion. It is composed of 35 to 85 % olivine and 10 to 50 % pyroxenes (orthopyroxene > clinopyroxene), plus minor Cr-spinel. In this type of rock, both orthopyroxene and clinopyroxene commonly occur as large oikocrysts enclosing small olivine crystals to form a poikilitic texture (Fig. 3a). In some places, lherzolite contains minor amounts of plagioclase.

Dunite is composed of 80–90 % olivine and 1–8 % pyroxenes, plus minor Cr-spinel which commonly occurs as small inclusions enclosed in olivine crystals (Fig. 3b). Olivine websterite is composed of 10–40 % olivine and 60–90 % pyroxenes.

The most abundant types of sulfide mineralization in the Jinchuan deposit are net-textured (Fig. 3c) and disseminated sulfides. Massive sulfides are rare in the deposit. The net-textured sulfide ores are most commonly associated with dunite (Fig. 3c), whereas the disseminated sulfide ores are most commonly associated with both dunite and lherzolite. The magmatic sulfide assemblage is composed of pyrrhotite, pentlandite, and chalcopyrite (Fig. 3d). The associated platinum-group minerals include native platinum, moncheite (PtTe_2), melonite ($\text{NiPdPt}(\text{TeBi})_2$), michenerite (PdBiTe), sperrylite (PtAs_2), and sudburyite (PtPdSd) (SGU 1984; Prichard et al. 2013).

The Jinchuan ultramafic rocks have been variably altered during postmagmatic hydrothermal overprinting. Olivine is partially replaced by serpentine plus secondary magnetite, pyroxenes are partially replaced by chlorite plus tremolite or actinolite, and Cr-spinel and magmatic sulfides are partially replaced by secondary magnetite.

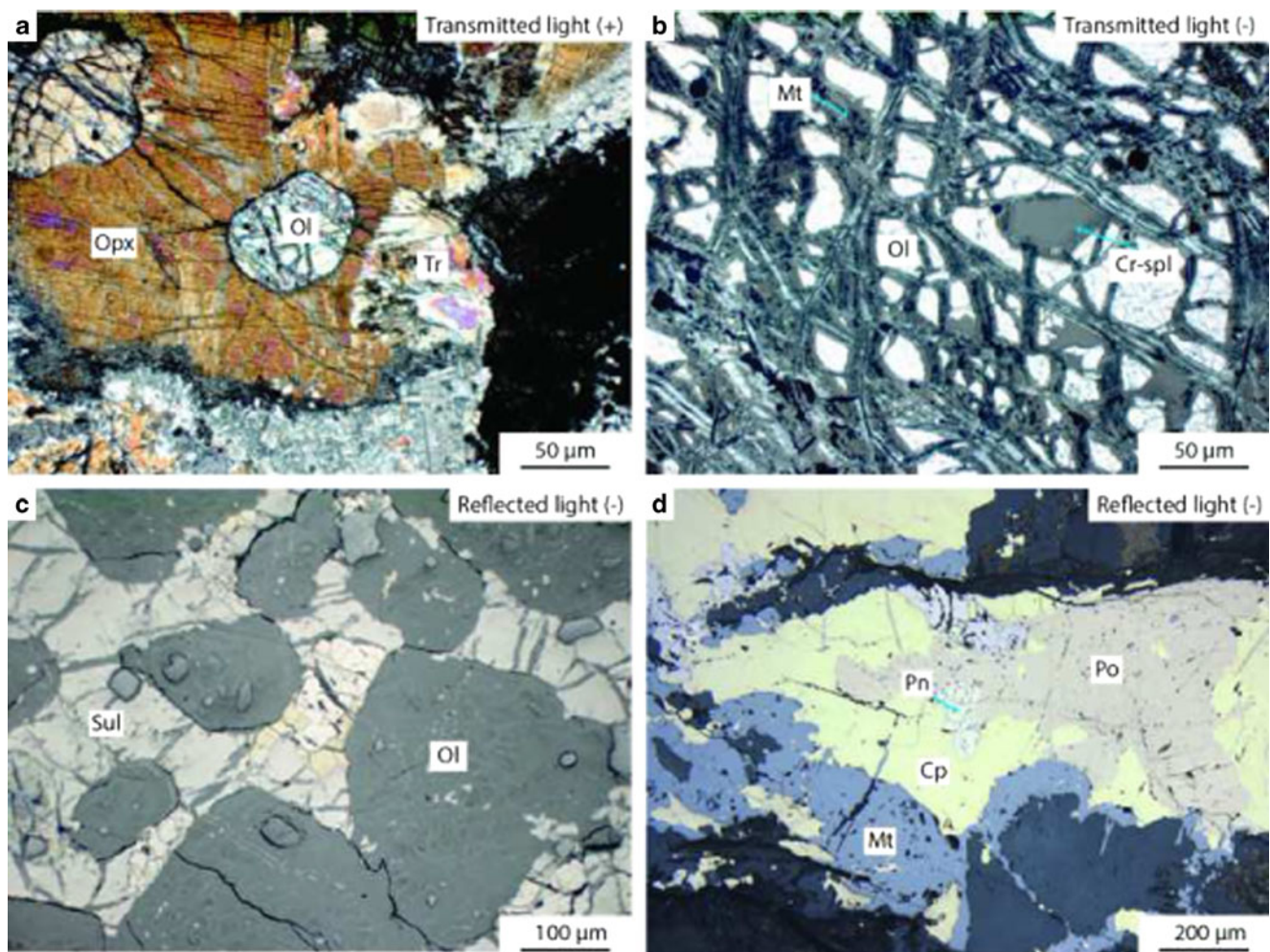


Fig. 3 Photomicrographs of lherzolite (a), dunite (b), and net-textured sulfide mineralization (c, d). Mineral abbreviations: *Cp* chalcopyrite, *Cr-spl* Cr-spinel, *Mt* magnetite, *Ol* olivine, *Opx* orthopyroxene, *Pn* pentlandite, *Po* pyrrhotite, *Sul* sulfide, *Tr* tremolite

Samples and analytical methods

Because the previously published data are for the upper part of the Jinchuan deposit, new samples used in this study were collected from greater depths of the deposit as currently oriented following tilting along underground tunnels at elevations between 978 and 1250 m, corresponding to ~500–750 m below the surface in several important segments of the intrusion: nine samples (L22-1 to L22-10) from I-24 at 1160 m elevation, seven samples (16-W1 to 16-E7) from I-6 at 1220 m elevation, eight samples (11-1 to 11-13) from II-10 at 978 m elevation, nine samples (1078-1 to 1078-10) from II-14 at 1078 m elevation, and eight samples (R25-1 to R25-10) from II-25 at 1250 m elevation (Fig. 2c). Our samples encompass all major ore types of the deposit and the most important rock types of the host intrusion.

Mineral compositions were determined by wavelength dispersive X-ray emission microanalysis using a CAMECA SX50 electron microprobe in the Department of Geological Sciences, Indiana University. The analytical conditions for major elements were 15 keV accelerating voltage, 20 nA beam current, 1 μm beam size, and 20 s peak counting time. International mineral standards were used for calibration.

Whole-rock major and trace element compositions were determined by X-ray fluorescence (XRF) and inductively coupled plasma-mass spectrometry (ICP-MS), respectively, in the MOE Key Laboratory of Western China Mineral Resources and Geological Engineering, Chang'an University, Xi'an, China. Rock powders were fused to make glass disks for XRF analysis and acid-digested in steel-jacketed Teflon "bombs" to produce solution for ICP-MS analysis.

The concentrations of Ni and Cu in sulfide-bearing samples were analyzed by X-ray fluorescence, and S concentrations in whole rocks were determined using a Vario EL2 elemental analyzer in the Xi'an Institute of Geology and Mineral Resources, China. The concentrations of PGE in sulfide-mineralized samples were determined by the combination of NiS bead pre-concentration and Te co-precipitation, followed by ICP-MS analysis at Guangzhou Institute of Geochemistry, Chinese Academy of Sciences. The concentrations of PGE in sulfide-poor samples were determined by Carius tube digestion isotope dilution, followed by ICP-MS analysis in the same laboratory. The detailed analytical procedures, blank concentrations, and detection limits are given in Sun et al. (2000) and Sun et al. (2009).

Whole-rock Sr-Nd isotope analyses were carried out using a MC-ICP-MS instrument in the State Key Laboratory of Continental Dynamics, Northwest University, Xi'an, China, following the procedures given in Zong et al. (2012). The total procedural blanks for Sr and Nd were 1.5 and 1.1 ng, respectively. The values for the La Jolla Nd standard and the SRM987 Sr standard measured together with samples were $^{143}\text{Nd}/^{144}\text{Nd} = 0.511861 \pm 10$ and $^{87}\text{Sr}/^{86}\text{Sr} = 0.710263 \pm 10$, respectively.

Zircon grains were separated from a ~50-kg lherzolite sample (JC-1) using standard mineral separation techniques. Euhedral, colorless, transparent, and undeformed zircon crystals were mounted in epoxy resin disk and then polished, cleaned, and carbon-coated. The cathodoluminescence and back-scattered electron images were used together to select the grains that have simple zoning patterns for Hf isotope analysis. The Hf isotopes of the selected zircon crystals from the Jinchuan ultramafic intrusion were carried out in situ using a Newwave UP213 laser-ablation microprobe, attached to a Neptune multicollector ICP-MS at MRL Key Laboratory of Metallogeny and Mineral Resource Assessment, Institute of Mineral Resources, Chinese Academy of Geological Sciences, Beijing. Instrumental conditions and data acquisition were the same as those given in Hou et al. (2007) and Wu et al. (2006). The laser beam size was about 40–55 μm . Zircon GJ1 standard (Morel et al. 2008) was used as reference.

Multiple S isotopes of sulfide minerals were determined in the Department of Geological Sciences, Indiana University. Sulfides were drilled under a microscope from polished sections using a 0.3-mm carbide drill bit. The sulfide separates were reacted with a Cr-reducing solution, liberating H_2S , which was then trapped using a zinc acetate solution and reacted with AgNO_3 to produce Ag_2S . The Ag_2S samples prepared for analysis using SF_6 were wrapped in Al foil and placed in Ni reaction vessels with excess F_2 derived from a He-F mixture, and heated to 450 °C overnight. The resulting SF_6 was condensed from the residual F_2 into a liquid-nitrogen cooled trap and subsequently distilled from the trap at 115 °C to the injection loop of a gas chromatograph (GC) at –177 °C. GC purification of SF_6 was undertaken using a 1/8-in. diameter, 12-ft-long Haysep column with a He carrier flow at 20 mL/min. The SF_6 peak was registered on a thermal conductivity detector and then isolated by freezing into a liquid-nitrogen cooled trap. Purified SF_6 was introduced into the Finnigan MAT 253 mass spectrometer and analyzed using a dual inlet method. International standards IAEA S-1 and S-2 were used to calibrate the SF_6 used as the working gas in the mass spectrometer. IAEA S-1 anchors the $\delta^{34}\text{S}$ scale at –0.3 ‰, with a $\Delta^{33}\text{S}$ of 0.10 ‰. IAEA S-2 has values in the IU lab of 22.33 ‰ $\delta^{34}\text{S}$ and 0.01 ‰ $\Delta^{33}\text{S}$. Uncertainties based on multiple measurements are ± 0.2 ‰ for $\delta^{34}\text{S}$ and ± 0.01 ‰ for $\Delta^{33}\text{S}$. All values are reported in standard delta notation with multiplication by 1000 to render values in ‰.

Analytical results

Mineral compositions

The average compositions of olivine crystals from the lower part of the Jinchuan ultramafic intrusion are provided as

electronic supplementary materials (ESM-1). These new data, together with previously published data, are used to illustrate the spatial variations of mineral compositions in the Jinchuan ultramafic intrusion (Fig. 4). The Fo contents of olivine crystals in the sulfide-poor samples (<0.5 wt% S) from the intrusion show no clear correlation with whole-rock Zr concentrations (Fig. 4a). In this type of rock, olivine crystals show a positive correlation between Ni and Fo contents (Fig. 4b). In contrast, olivine crystals in the sulfide-bearing samples show a negative correlation between Ni and Fo contents for individual ore zones (Fig. 4c). Among the different ore zones, olivine crystals from the westernmost part of segment II (i.e., II-W-a, Fig. 2c) have the lowest Fo and Ni contents. This difference is coupled with different Cr-spinel compositions for the different zones. As shown in Fig. 4d, Cr-spinel crystals from segment II-W-a tend to have lower $\text{Cr}/(\text{Cr}+\text{Al})$ and higher $\text{Fe}^{3+}/(\text{Cr}+\text{Al}+\text{Fe}^{3+})$ than those from the other segments.

Whole-rock major and trace elements

The concentrations of major and trace elements in the samples from the lower part of the Jinchuan ultramafic intrusion are provided as electronic supplementary materials (ESM-2). These new data, together with previously published data, are

used to illustrate the variations of $\text{Mg}^\#$ [$100\text{MgO}/(\text{MgO}+\text{FeO}^\text{T})$, molar] and loss on ignition (LOI) in sulfide-poor samples (<0.5 wt% S) from different parts of the intrusion (Fig. 5). The amount of LOI mainly reflects the degree of serpentinization, because these rocks are olivine cumulates and serpentine is the most important hydrous mineral replacing a large proportion of olivine in the samples. It is important to note that the samples from II-W-a (Fig. 2b), which have rather unique olivine and Cr-spinel compositions (Fig. 4), do not have higher LOI than those from elsewhere in the intrusion (Fig. 5). **This observation indicates that the somewhat distinct mineral compositions in II-W-a are a primary signature, not due to higher degrees of hydrothermal alteration.**

Chalcophile elements

The concentrations of Ni, Cu, S, and PGE in the sulfide-mineralized samples from the lower part of the Jinchuan deposit are given in Table 1. These new data, together with previously published data, are used to examine how the chalcophile elements vary with each other and in space. In the following presentation, the rare samples **containing >15 wt% S (i.e., semimassive and massive sulfide ores) as well as sulfide-poor samples containing <1 wt% S are**

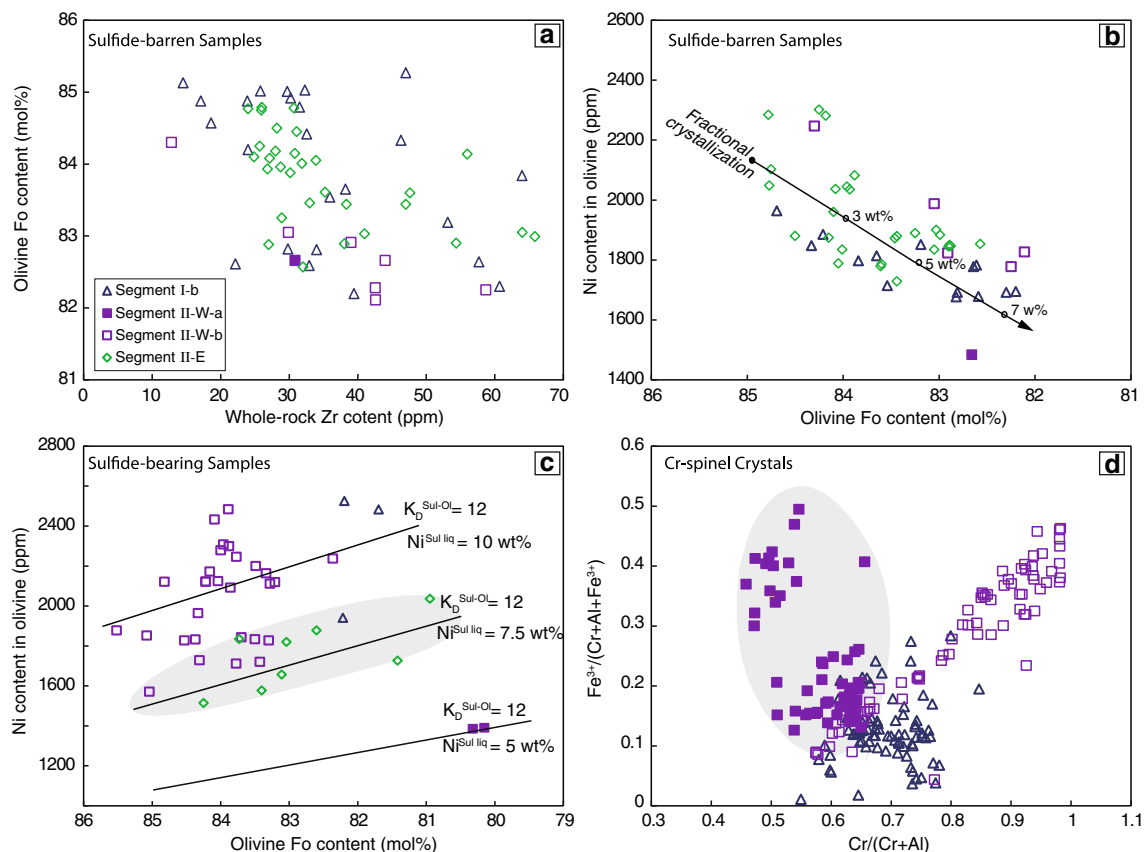


Fig. 4 Compositional variations of different types of olivine (a–c) and Cr-spinel (d). Data are from this study, Barnes and Tang (1999), Chai and Naldrett (1992a), Chen et al. (2009), de Waal et al. (2004), and Li et al. (2004). See text for the modeling

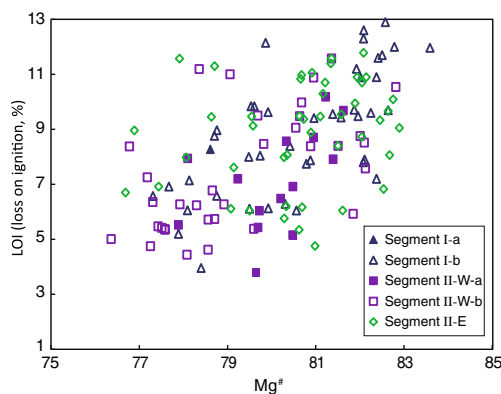


Fig. 5 Plot of whole-rock $Mg^{\#}$ [$100MgO/(MgO+FeO^T)$, molar] versus loss on ignition (LOI). Data are from this study, Song et al. (2006, 2009, 2012), Tonnelier (2010), de Waal et al. (2004), and Chai and Naldrett (1992a)

excluded. The selected samples represent typical disseminated and net-textured sulfides ores in the deposit. In these samples,

positive correlations exist between Ni and S (Fig. 6a) and between Pd and S (Fig. 6c). In contrast, Cu and Ir do not show any correlation with S abundance in the samples (Fig. 6b, d). The lack of correlation between Cu and S contents in the samples is not really surprising, because chalcopyrite fractionation is common in magmatic sulfide deposits. A closer examination reveals that the lack of correlation between Ir and S is caused by Ir depletion in the samples from II-W-a and its neighboring zone to the west, zone I-a (Fig. 2b). Excluding these Ir-depleted samples, the rest of the samples in fact show a weak positive correlation between Ir and S (Fig. 6d). It is important to note that except one sample, the Ir-depleted samples do not show Pd enrichment (Fig. 6c).

The concentrations of chalcophile elements recalculated to those in 100 % sulfide are commonly referred to as metal tenors in the literature. We have used the equation of Barnes and Lightfoot (2005), which is based on the assumption that the magmatic sulfide assemblage is composed of pyrrhotite,

Table 1 Concentrations of chalcophile elements in different parts of the Jinchuan deposit

Segment	Sample	S (wt%)	Os (ppb)	Ir (ppb)	Ru (ppb)	Rh (ppb)	Pt (ppb)	Pd (ppb)	Ni (wt%)	Cu (wt%)
Segment I-a	16-W5	9.37	0.08	0.08	0.10	0.07	142	232	0.86	6.39
	16-W7	10.7	0.31	0.29	0.14	0.66	455	303	2.33	5.67
	16-E7	7.02	0.34	0.22	0.15	0.11	493	348	2.73	2.24
Segment I-b	16-W1	8.62	66.0	67.7	56.0	20.0	22.1	116	2.00	0.61
	16-W3	7.77	42.4	20.7	8.85	20.8	6571	210	2.89	1.80
	16-E5	11.5	68.8	69.2	54.3	24.7	21.1	567	4.07	1.64
	L22-3	11.6	28.7	33.1	23.0	16.8	11,265	569	3.62	0.28
	L22-4	16.1	98.4	96.3	83.2	44.7	55.3	354	3.26	4.11
	L22-9	1.25	8.02	7.21	4.43	2.46	53.7	38.8	0.41	0.14
	L22-10	1.96	23.3	6.27	4.53	2.96	41.5	31.5	0.48	0.27
Segment II-W-a	11-1	7.57	0.78	0.37	0.42	0.46	83.9	46.9	2.80	2.73
	11-7	5.97	0.80	0.68	0.60	1.18	239	254	3.26	0.92
	11-3	7.44	1.51	0.91	0.64	1.55	25,192	166	2.82	2.48
Segment II-W-b	11-5	9.41	64.7	71.1	56.0	27.8	16.6	324	2.94	0.50
	11-4	7.80	32.8	62.5	29.5	12.6	128	354	2.72	0.97
	11-13	10.2	89.6	89.3	72.7	29.9	23.9	377	3.04	1.77
	1078-6	8.49	2.64	2.37	0.56	3.69	101	147	2.64	1.20
	1078-7	6.96	11.7	7.61	1.83	5.31	317	19.7	1.72	2.22
	1078-5	4.56	7.10	6.81	3.29	3.62	69.1	77.6	1.11	0.95
	1078-8	6.03	24.2	20.4	14.8	10.4	119	133	1.78	0.78
	1078-9	7.61	40.6	42.0	34.6	16.3	33.6	59.7	1.93	0.94
	1078-2	7.90	60.3	57.9	46.4	15.7	83.5	93.2	2.30	0.41
	1078-1	7.48	62.8	57.4	43.6	23.3	15.4	54.2	1.45	0.70
	R25-1	2.38	1.73	1.06	0.94	0.69	22.7	15.0	0.54	0.52
	R25-2	3.02	3.15	1.26	1.41	0.95	17.9	13.9	0.60	0.81
	R25-3	2.47	3.92	2.10	1.98	1.41	19.1	12.4	0.77	0.14
	R25-5	2.69	5.03	2.89	2.77	1.91	29.4	38.0	0.63	0.97
	R25-6	2.23	6.28	2.71	3.03	1.81	49.6	11.8	0.73	0.11
	R25-7	1.58	1.36	0.56	0.53	0.39	7.50	9.95	0.43	0.17

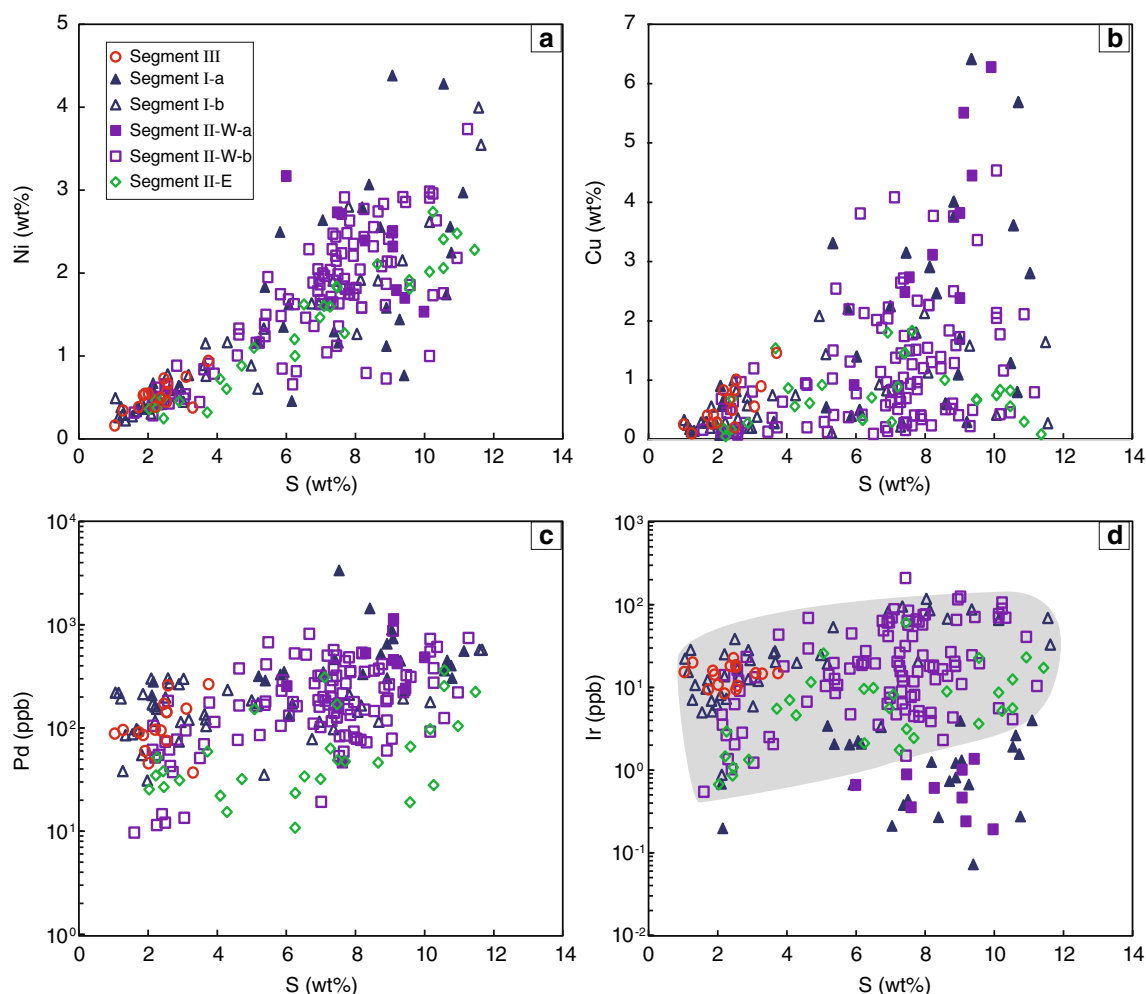


Fig. 6 a–d Plots of metal versus S contents in whole rocks. Data are from this study, Chai and Naldrett (1992a, b), Chen et al. (2013), Song et al. (2006, 2009), Su et al. (2008), and Tonnelier (2010)

pentlandite, and chalcopyrite (Naldrett and Duke 1980; Li et al. 2001), to calculate the metal tenors in the Jinchuan deposit. The amounts of Ni from silicates in the samples were corrected using the silicate proportion in the samples and the average content of Ni in the sulfide-poor samples (<0.1 wt% S) in the Jinchuan intrusion. The results show a clear positive correlation between Rh, Ru, and Ir (Fig. 7a, b) and a weak positive correlation between Pd and Pt (Fig. 7c). In contrast, no correlation is present between Pd, Cu, and Ir (Fig. 7d–f). Zone II-W-a and its neighboring zone to the west, zone I-a (Fig. 2b), are clearly depleted in Ir relative to Pd and Cu (Fig. 7d, e).

The primitive mantle-normalized patterns of chalcophile metal tenors in the different parts of the Jinchuan deposit are illustrated in Fig. 8. The bulk sulfide ores in the westernmost part of the deposit (segment III, Fig. 2b) are characterized by the least fractionated patterns and minor positive Pt anomaly (Fig. 8a). Their Pd/Ir ratios are similar to those of picrites from the Emeishan flood basalt province in SW China and northern Vietnam (Li et al. 2012). Compared to segment III, the bulk

sulfide ores in segment I-b, which occur to the east (Fig. 2b), have similar chalcophile element patterns to those of segment III but also show a negative Pt anomaly in addition to a positive Pt anomaly (Fig. 8b). Farther to the east occur two ore zones (I-a, II-W-a, Fig. 2b) that are significantly depleted in Ir, Ru, and Rh (Fig. 8c, d). No significant IPGE depletions are observed in the eastern parts of the deposit (Fig. 2b) such as II-W-b (Fig. 8e) and II-E (Fig. 8f). Although these two ore zones have similar PGE patterns, the PGE tenors in the latter tend to be lower.

Hf-Sr-Nd isotopes

The Hf isotopic compositions of 22 zircon crystals separated from a large lherzolite sample from the Jinchuan intrusion are listed in Table 2. The calculated ε_{Hf} ($t=831$ Ma) values of zircons vary between -4 and -7 (Fig. 9a), which are similar to the previous results (Jiao et al. 2014; Tang et al. 2014).

Whole-rock Sr-Nd isotopic compositions of the Jinchuan intrusion are listed in Table 3. The calculated ε_{Nd} ($t=831$ Ma)

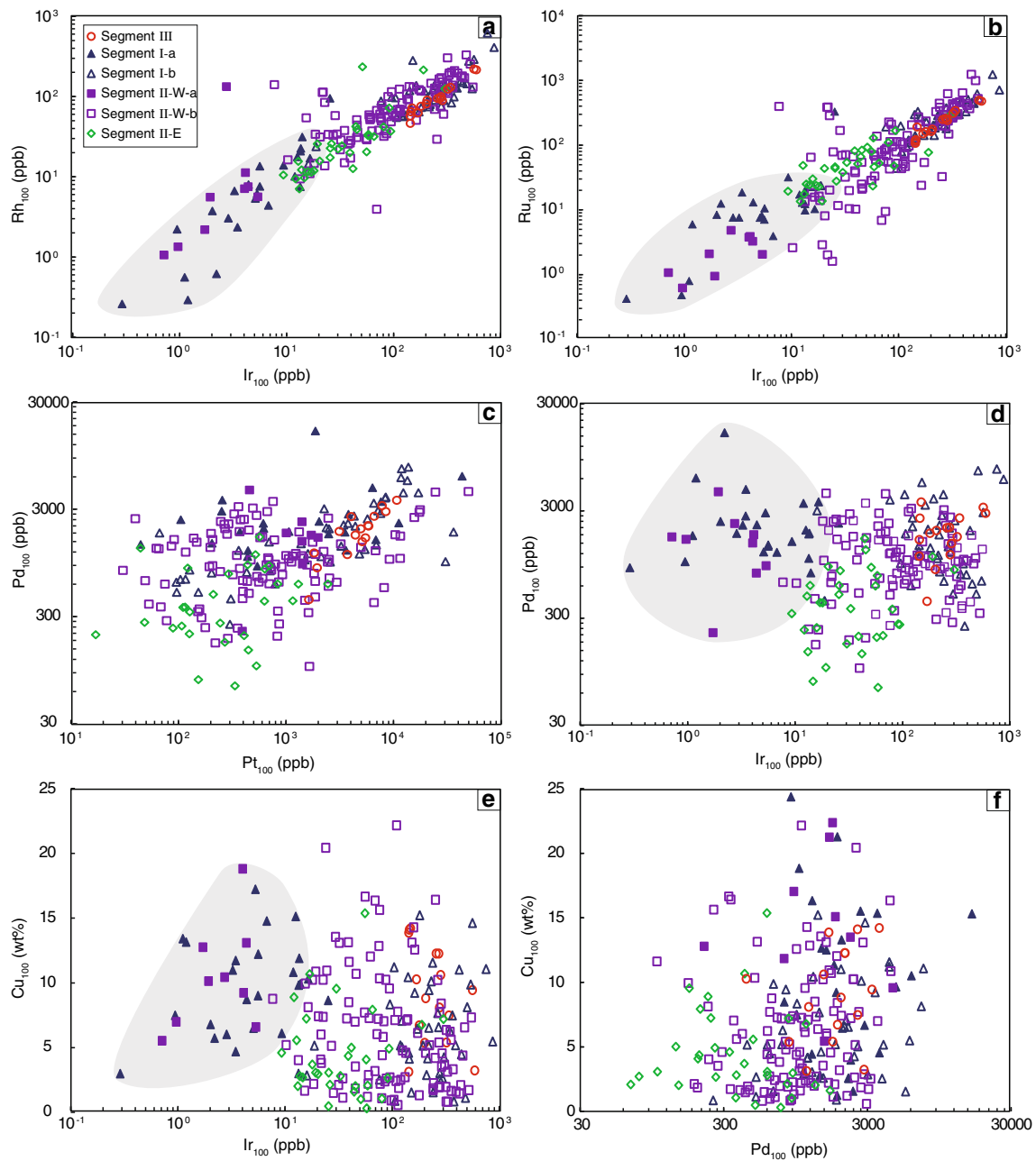


Fig. 7 a–f Covariations of metal tenors (i.e., in recalculated 100 % sulfide). Data are from this study, Chai and Naldrett (1992a, b), Chen et al. (2013), Song et al. (2006, 2009), Su et al. (2008), and Tonnellerie (2010)

values and initial $^{87}\text{Sr}/^{86}\text{Sr}$ ratios vary from -9.2 to -10.5 and from 0.7077 to 0.7093 , respectively. These new data, together with the previous results (Tang et al. 1992; Zhang et al. 2004), form a positive correlation, pointing toward the upper crust (Fig. 9b).

Multiple S isotopes

The multiple S isotopic compositions of the Jinchuan sulfide ores are given in Table 4. Among a total of nine Jinchuan samples analyzed by us, only one of them shows slightly more

elevated $\delta^{34}\text{S}$ value (2.3‰) than the MORB mantle values (-1.57 to 0.6‰ , Labidi et al. 2013, 2014). A comparison with a magmatic sulfide deposit in the **midcontinent rift system**, USA, which shows anomalous $\Delta^{33}\text{S}$ values (Ding et al. 2012) and Archean sedimentary rocks worldwide (Johnston 2012) is illustrated in Fig. 10. Four of the samples from the Jinchuan deposit show anomalous $\Delta^{33}\text{S}$ values varying from 0.12 to 2.67‰ . The sample with a $\delta^{34}\text{S}$ value of 2.3‰ does not have an anomalous $\Delta^{33}\text{S}$ value and is consistent with a post-Archean crustal source (Ripley and Li 2007). Samples with anomalous $\Delta^{33}\text{S}$ values are indicative of a contribution

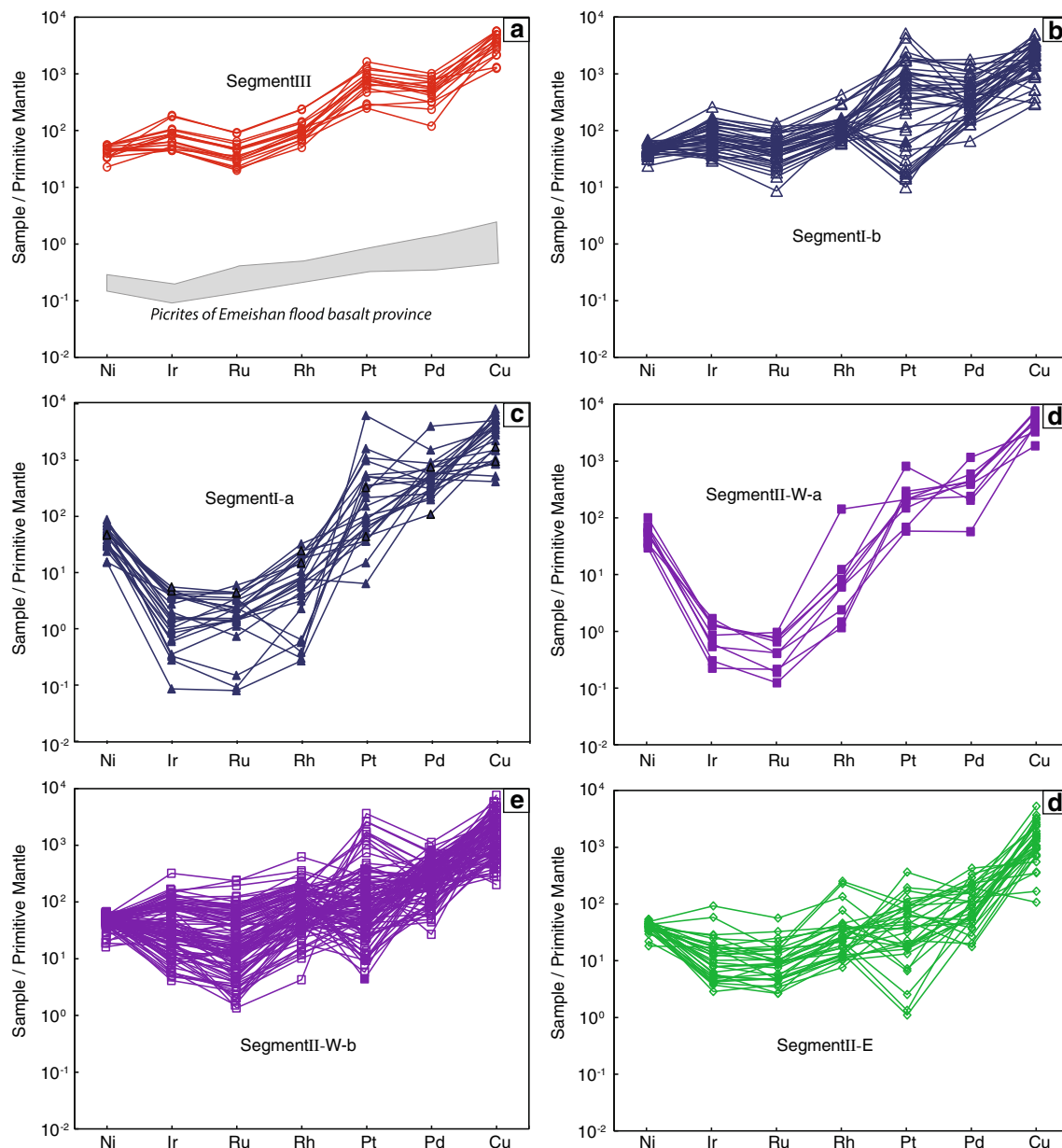


Fig. 8 a–f Primitive mantle-normalized patterns of metal tenors in the different parts of the Jinchuan deposit. Data for the deposit are from this study, Chai and Naldrett (1992a, b), Chen et al. (2013), Song et al. (2006, 2009), Su et al. (2008), and Tonnellier (2010). The picrite data are from Li et al. (2012)

of S from Archean crustal rocks (Farquhar and Wing 2003; Fiorentini et al. 2012).

Modeling and discussion

Oxidation state

The oxidation state of a magmatic sulfide system can be estimated from Ni concentration in sulfide liquid and the exchange coefficient for Fe and Ni between olivine and sulfide liquid, $K_D [(X_{NiS}/X_{FeS})^{sulfide\ liquid}/(X_{NiS}/X_{FeS})^{olivine}]$, molar]

(Brenan 2003). Recently, Barnes et al. (2013) used available experimental data to calibrate such a relationship. We have used the equation of Barnes et al. (2013) to estimate the oxidation state of the Jinchuan magmatic sulfide system using our new data. The results are illustrated in Fig. 11. Based on sulfide-olivine textures, Tonnellier (2010) suggested that magmas which produced segment II-W were more oxidized than those that gave rise to segments I and II-E. The results from this study indicate that the oxidation states of the different segments are similar, varying within a small range between quartz-fayalite-magnetite (QFM)+0.3 and QFM+1 (i.e., fO_2

Table 2 Lu–Hf isotopes of zircon crystals from the Jinchuan deposit

Grain	Age, Ma	$^{176}\text{Yb}/^{177}\text{Hf}$	2σ	$^{176}\text{Lu}/^{177}\text{Hf}$	2σ	$^{176}\text{Hf}/^{177}\text{Hf}$	2σ	$(^{176}\text{Hf}/^{177}\text{Hf})_i$	ε_{Hf} (t)
1	845	0.022781	0.000167	0.000767	0.000005	0.282110	0.000012	0.282098	−5.2
2	842	0.027200	0.000195	0.000828	0.000007	0.282147	0.000013	0.282134	−4.0
3	842	0.047155	0.000271	0.001483	0.000009	0.282107	0.000014	0.282083	−5.8
4	840	0.054819	0.000161	0.001732	0.000005	0.282072	0.000013	0.282045	−7.2
5	838	0.028303	0.000062	0.000799	0.000002	0.282084	0.000015	0.282071	−6.3
6	836	0.041559	0.000042	0.001337	0.000001	0.282130	0.000015	0.282109	−5.0
7	837	0.042504	0.000054	0.001293	0.000001	0.282131	0.000014	0.282111	−4.9
8	843	0.017426	0.000111	0.000455	0.000002	0.282115	0.000014	0.282108	−4.9
9	830	0.035254	0.000039	0.001140	0.000002	0.282134	0.000016	0.282117	−4.9
10	836	0.036156	0.000036	0.001140	0.000001	0.282154	0.000016	0.282136	−4.1
11	830	0.042112	0.000189	0.001395	0.000006	0.282130	0.000015	0.282108	−5.2
12	844	0.040642	0.000212	0.001311	0.000006	0.282122	0.000017	0.282101	−5.1
13	845	0.029465	0.000044	0.000932	0.000001	0.282107	0.000014	0.282093	−5.4
14	837	0.069429	0.000118	0.002203	0.000002	0.282131	0.000016	0.282096	−5.4
15	833	0.015168	0.000152	0.000427	0.000004	0.282123	0.000011	0.282117	−4.8
16	835	0.027853	0.000015	0.000875	0.000001	0.282126	0.000012	0.282112	−4.9
17	833	0.063543	0.000303	0.002014	0.000009	0.282079	0.000015	0.282047	−7.2
18	846	0.027528	0.000137	0.000807	0.000004	0.282130	0.000013	0.282117	−4.5
19	833	0.062971	0.000130	0.001885	0.000004	0.282201	0.000013	0.282172	−2.8
20	840	0.033078	0.000277	0.001033	0.000008	0.282146	0.000016	0.282130	−4.2
21	846	0.033533	0.000099	0.000994	0.000003	0.282117	0.000013	0.282102	−5.1
22	835	0.028105	0.000196	0.000849	0.000005	0.282108	0.000012	0.282095	−5.5
23	832	0.025722	0.000020	0.000785	0.000001	0.282133	0.000015	0.282121	−4.7

0.3–1 log unit higher than the value for the quartz-fayalite-magnetite buffer). Compared with the Voisey's Bay deposit in Labrador, Canada, which represents the most reduced magmatic sulfide deposit known world-wide (Barnes et al. 2013), the magmas that produced the Jinchuan deposit were more oxidized.

Controls on PGE tenor variations

Positive correlations between IPGE (Ir, Os, Ru, Rh) or between PPGE (Pt, Pd) in the Jinchuan deposit (Fig. 7a–c) are expected because each of these groups behaves similarly during immiscible sulfide segregation from magma (Mungall and

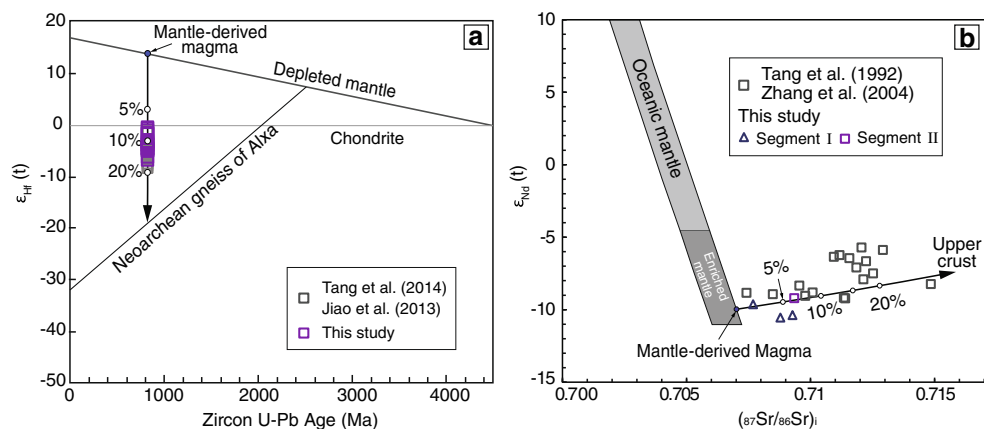


Fig. 9 Zircon Hf isotopes (a) and whole-rock Sr–Nd isotopes (b) of the Jinchuan intrusion. The age of gneiss in southern Alxa is from Gong et al. (2012). The Hf isotope evolution of the gneiss is calculated using the parameters for the upper crust. Parameters used in mixing calculations: 320 ppm Sr, 27 ppm Nd, 7.8 Hf, $(^{87}\text{Sr}/^{86}\text{Sr})_i=0.72$, $\varepsilon_{\text{Nd}}=-6$, and ε_{Hf}

(830 Ma)=−19 for the upper crust; 100 ppm Sr, 10 ppm Nd, 0.8 Hf, $(^{87}\text{Sr}/^{86}\text{Sr})_i=0.707$, $\varepsilon_{\text{Nd}}=-10$, and $\varepsilon_{\text{Hf}}=14$ for the mantle-derived magma. The element concentrations in the end members are from Rudnick et al. (2003) and Sun and McDonough (1989)

Table 3 Sr and Nd isotopes of the Jinchuan deposit

Sample	Segment	Concentration (ppm)				Isotope ratio					
		Rb	Sr	Sm	Nd	$^{87}\text{Rb}/^{86}\text{Sr}$	$^{87}\text{Sr}/^{86}\text{Sr}$	$(^{87}\text{Sr}/^{86}\text{Sr})_i$	$^{147}\text{Sm}/^{144}\text{Nd}$	$^{143}\text{Nd}/^{144}\text{Nd}$	$\varepsilon_{\text{Nd}}(t)$
L22-6	I	71.75	400.26	4.43	20.13	0.5190	0.715428	0.709275	0.1340	0.511766	−10.37
L22-7	I	39.53	116.70	3.39	14.78	0.9808	0.719310	0.707681	0.1395	0.511834	−9.62
L22-8	I	21.15	116.59	2.04	8.74	0.5253	0.715010	0.708782	0.1420	0.511801	−10.54
R25-4	II-W	9.89	75.01	0.85	3.55	0.3819	0.713868	0.709340	0.1451	0.511886	−9.20

Brenan 2014) and subsequent fractional crystallization of monosulfide solid solution (MSS) from sulfide liquid (Li et al. 1996). A lack of correlation between PPGE and IPGE in the Jinchuan deposit (Fig. 7d) is also expected because the two different groups behave differently during MSS fractional crystallization (Li et al. 1996). Thus, the effects of these two different processes can be evaluated using a plot of PPGE versus IPGE (Li and Naldrett 1994). Here we use the plot of Pd versus Ir to model their effects on PGE tenor variations in the Jinchuan deposit (Fig. 12). Platinum was not selected for this type of modeling because some samples from the Jinchuan deposit exhibit pronounced positive or negative anomalies (Fig. 8). Iridium was chosen to represent IPGE because it has the lowest detection limit. Three samples with unusual Pd/Ir ratios (i.e., far too low or far too high compared with the rest of the samples) are excluded from Fig. 12. Our modeling results show that except I-a and II-W-a, the PGE tenor variations in the other ore zones of the deposit can be explained by variable R-factors (magma/sulfide mass ratio, Campbell and Naldrett 1979) plus moderate degrees of sulfide liquid fractionation on cooling. The modeling results show that the estimated R-factors for the bulk sulfide ores in segment II-E are much lower than those in segments I-b and II-W-b (Fig. 12). This observation is consistent with lower Ni content in the sulfide liquid of this zone relative to that of the other

two zones inferred from olivine Fo-Ni relations (Fig. 4c). Given that the effect of hydrothermal alteration cannot be completely eliminated by the choice of elements, the modeling results are remarkable.

As shown in Fig. 12, a different parental magma with lower Pd/Ir than that for the other ore zones of the Jinchuan deposit is required to explain the two spatially closely associated, IPGE-depleted ore zones, I-a and II-W-a in the western part of the deposit (Fig. 2b). This explanation is supported by the fact that the depletions of IPGE in these zones are not coupled to enrichments in Pd and Cu (Fig. 7d, e), which rules out MSS fractional crystallization as the main cause for the observed IPGE depletions in these zones. Another possibility for the unusual PGE metal tenors in these ore zones is postmagmatic hydrothermal overprinting that added some Pd and Pt to these two zones. However, mineralogical and petrological evidence to support or negate this interpretation has not been found. As shown in Fig. 5, the degrees of serpentinization, as inferred from whole-rock LOI values, are not higher in these two zones than in the other parts of the deposit. Thus, a distinct IPGE-depleted parental magma for these two spatially closely related ore zones remains the best choice. We suggest that the depletion of IPGE in the magma resulted from fractional crystallization of Cr-spinel and olivine, which is consistent with the observation that olivine and Cr-spinel are generally more

Table 4 Multiple sulfur isotopes of the Jinchuan deposit

Sample	Segment	S (wt%)	Sulfide mineral assemblage	$\delta^{33}\text{S}$	$\delta^{34}\text{S}$	$\Delta^{33}\text{S}$
16-E3	I-a	2.18	Po+Cp+Pn	0.35	0.38	0.15
050712-4	I-a	2.08	Po+Cp+Pn	−0.49	−1.19	0.12
L22-4	I-b	2.00	Po+Cp+Pn	1.17	2.30	−0.01
1078-2	II-W-b	2.27	Po+Cp+Pn	−0.32	−0.76	0.07
050713-5	II-W-b	2.16	Po+Cp+Pn	−0.13	−0.24	0.00
050713-6	II-W-b	2.13	Po+Cp+Pn	2.72	0.10	2.67
R25-1	II-W-b	1.98	Po+Cp+Pn	−0.25	−0.52	0.02
R25-3	II-W-b	1.03	Po+Cp+Pn	−0.36	−1.12	0.22
R25-7	II-W-b	2.14	Po+Cp+Pn	−0.57	−1.12	0.00

Po pyrrhotite, Cp chalcopyrite, Pn pentlandite

$\Delta^{33}\text{S} = \delta^{33}\text{S} - 0.515 * \delta^{34}\text{S}$

$\delta^{33}\text{S}$, $\delta^{34}\text{S}$, $\Delta^{33}\text{S}$ are in per mil (‰), relative to V-CDT.

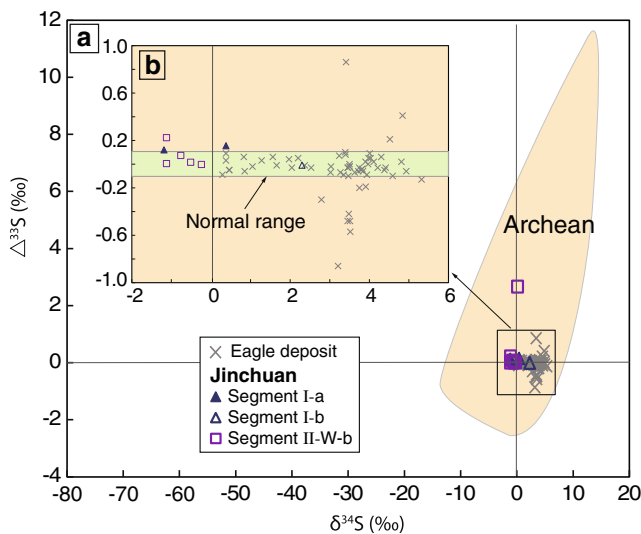


Fig. 10 a, b Multiple sulfur isotope systematics for the Jinchuan deposit. The Archean multiple sulfur isotopic data are from Johnston (2012). The values for the Eagle deposit are from Ding et al. (2012)

fractionated in the IPGE-depleted ore zones than elsewhere in the deposit (Fig. 4) and that IPGE are compatible whereas PPGE are incompatible in these two minerals in experimental basaltic systems (Brenan et al. 2003, 2012) and natural komatiites (Fiorentini et al. 2010). No olivine data are currently available for the sulfide-bearing samples from zone I-a, but available olivine data for the sulfide-bearing samples from the other IPGE-depleted zone (II-W-a) indicate significant Ni depletion in the coexisting sulfide liquid (Fig. 4c). This observation, together with lower Fo contents of olivine crystals from this zone, provides additional support for the above interpretation.

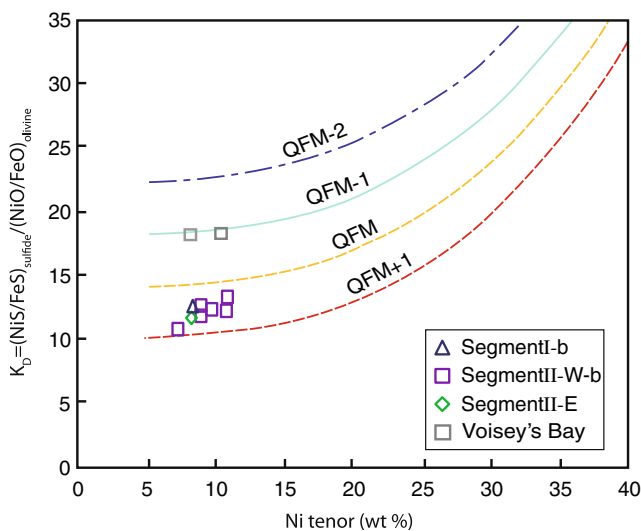


Fig. 11 The oxidation states of the Jinchuan magmatic sulfide deposit determined using olivine compositions and Ni contents in bulk sulfide using the method of Barnes et al. (2013). The values for the Voisey's Bay deposit are from Barnes et al. (2013)

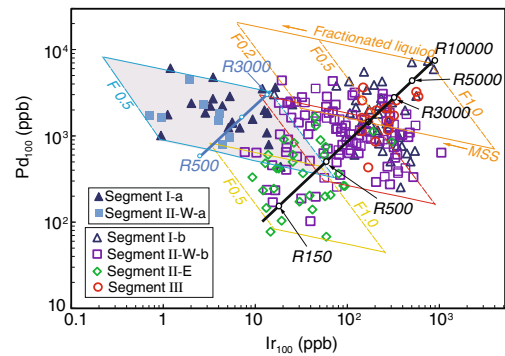


Fig. 12 Modeling of Pd and Ir tenor variations as the results of variable R-factors and MSS (monosulfide solid solution) fractional crystallization. Data are from this study, Chai and Naldrett (1992a, b), Chen et al. (2013), Song et al. (2006, 2009), Su et al. (2008), and Tonnelier (2010). Modeling parameters: sulfide/magma $D_{Ir}=30,000$, $D_{Pd}=30,000$; MSS/liquid $D_{Ir}=5$, $D_{Pd}=0.1$; initial magma Ir=0.005 ppb and Pd=1.1 ppb for I-a and II-W-a; initial magma Ir=0.12 ppb and Pd=1 ppb for the other zones. The selected D values between MSS and sulfide liquid and between sulfide liquid and magma are within the ranges of experimental values given by Li et al. (1996) and Mungall and Brenan (2014), respectively

Pronounced negative and positive Pt anomalies relative to Rh and Pd in some samples from the Jinchuan deposit (Fig. 8) may have resulted from Pt-mineral fractionation during sulfide liquid solidification on cooling (Prichard et al. 2013) or from postmagmatic hydrothermal alteration, or both (Li and Ripley 2011).

The role of fractional crystallization in sulfide saturation

We have used the MELTS program of Ghiorso and Sack (1995) and Asimow and Ghiorso (1998) to simulate the fractional crystallization of the primary magma for the Jinchuan intrusion given by Li and Ripley (2011) at 2 kb total pressure and oxidation state of QFM+0.5. The assumed oxidation state is close to the average value estimated from the olivine-sulfide Fe-Ni exchange reaction (Fig. 11). The variations of olivine Fo and Ni contents with the degree of fractional crystallization were obtained from the MELTS simulation and numerical modeling using the Rayleigh fractionation equation for trace elements, respectively, following the procedures given in Li et al. (2007). The partition coefficient of Ni between olivine and magma was calculated using the composition- and temperature-dependent equation of Li and Ripley (2010). The partition coefficients of Ni between clinopyroxene, plagioclase, and magma were assumed to be 1 and 0, respectively. The modeling results show that the parental magma of the Jinchuan intrusion formed by ~18 % fractional crystallization of the primary magma (Fig. 13) and that ~7 % additional fractional crystallization adequately explains the observed olivine Fo-Ni variations in the Jinchuan intrusion (Fig. 4b).

According to Naldrett (2011), who used pMELTS (Ghiorso et al. 2002) to determine the relationship between magma MgO composition and the degree of mantle partial melting,

a primary magma containing 18.5 wt% MgO (the same as the value for the Jinchuan primary magma, Li and Ripley 2011) requires ~18 wt% partial melting of a typical peridotite mantle. If the content of S in the source mantle was 200 ppm, as suggested by Palme and O'Neill (2014), the maximum amount of S in the primary magma would have been 1111 ppm if no sulfide was retained in the mantle. The temperature of the primary magma in the source mantle is estimated to be 1525 °C using the equation of Herzberg and Asimow (2015). Accordingly, the sulfur content at sulfide saturation (SCSS) in the primary magma in the source mantle is estimated to be ~2000 ppm using the equation of Li and Ripley (2009) (Fig. 13). A comparison between these two values reveals that such a primary magma could have retained the maximum S value of 1111 ppm when it left the source mantle. Sulfur is incompatible in Cr-spinel and silicate minerals crystallizing from basaltic magma. As a result, it will become progressively enriched in the residual magma during fractional crystallization. This is commonly referred to as residual enrichment, a process that may eventually induce sulfide saturation in magma (Ripley and Li 2013). However, in the case of Jinchuan, our modeling results show that ~33 wt% fractional crystallization is required for the residual magma to become sulfide-saturated at the crustal level (i.e., 2 kb) if no other processes were involved (Fig. 13). ~~A comparison between the modeling results and the observed Fo contents of olivine in the Jinchuan intrusion indicates that~~ sulfide saturation in the Jinchuan magmatic system took place before the primary magma had undergone ~18 wt% fractional crystallization, which indicates that fractional crystallization did not play a major role in triggering sulfide saturation in the Jinchuan magma (Fig. 13).

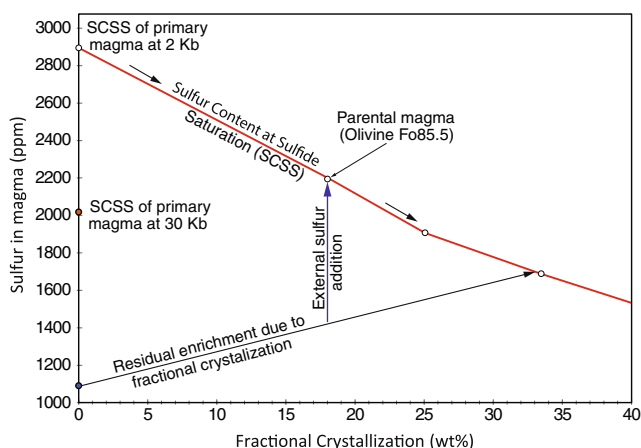


Fig. 13 Changes of sulfur contents in the Jinchuan magma due to fractional crystallization and external sulfur addition. Mineral abbreviations are the same as in Fig. 3. See text for modeling and explanations

Crustal contamination and external sulfur

Mixing calculations based on zircon Hf isotopes (Fig. 9a) and whole-rock Sr-Nd isotopes (Fig. 9b) indicate that the Jinchuan magma experienced up to ~20 % contamination with the bulk upper crust. The estimation based on Hf isotopes represents the maximum value, because the parental magma could be less depleted isotopically than the mantle end member used in the model (Fig. 9a). An alternative explanation for the low ε_{Nd} values of the Jinchuan intrusion (Fig. 9b) is the previous contamination with the lower crust characterized by extremely low ε_{Nd} values. Multiple S isotopes provide clear evidence that both Archean (elevated $\Delta^{33}\text{S}$ up to 2.67‰, Fig. 10) and post-Archean (elevated $\delta^{34}\text{S}$ up to 8‰, Ripley et al. 2005) crustal sulfur were involved in the generation of the Jinchuan deposit. Previously reported Re-Os isotope data for the Jinchuan deposit (γOs from 30 to 320; Yang et al. 2005, 2008; Zhang et al. 2004) are consistent with the S isotope data. Based on olivine composition and modeling results, the addition of external sulfur to the Jinchuan magma is inferred to have occurred before the magma had experienced ~18 wt% fractional crystallization (Fig. 13).

Previous sulfide segregation

Depending on the choice of R-factors, the estimated initial Pd content in the Jinchuan parental magma varies from ~0.3 ppb (Song et al. 2009) to ~3 ppb (Li and Ripley 2011). The former is based on an assumed minimum R-factor close to 1000, whereas the latter is based on an assumed minimum R-factor close to ~100. In our new model, a minimum R-factor of ~150 is used (Fig. 12). This value is similar to the value expected for a parental magma containing ~2300 ppm S at SCSS (Fig. 13), if sulfide segregation took place in a closed system, i.e., all the dissolved sulfur eventually segregated in situ from the magma as immiscible sulfide liquid on cooling. In an open system such as a flow-through system, only some of the sulfur in the magma could form immiscible sulfide liquid at a given location, resulting in a higher R-factor. The R-factor could increase further if the sulfide reacted with new, sulfide-unsaturated magma. On the other hand, the minimum R-factor could be lower if external sulfur was involved. Thus, the minimum R-factor in our model seems reasonable (Fig. 12). Accordingly, the initial content of Pd in the parental magma of the Jinchuan deposit is estimated to be ~1 ppb. This value is only about 1/10 of the average value for PGE-undepleted picrites associated with continental flood basalts (Li et al. 2012; Lightfoot and Keays 2005), indicating that the Jinchuan parental magma indeed experienced previous sulfide segregation, as suggested previously by some researchers (e.g., Song et al. 2009; Li and Ripley 2011). Since the primary magma of the Jinchuan deposit was not saturated in sulfide when it left the source magma (Fig. 13), this event is believed

to have occurred in the staging magma chamber or during magma ascent.

A dynamic ore genesis model

Based on data available to date, we have proposed a self-consistent, dynamic ore genesis model for the Jinchuan deposit (Fig. 14). Due to the lack of PGE data, segment III, the westernmost part of the intrusion, and segment IV, the easternmost part of the intrusion (Fig. 2a), are excluded from our model. In this model, the eastward increase of PGE tenors in the deposit, excluding two relatively small ore zones, I-a and II-W-a (Figs. 2b and 12), is explained by metal upgrading by a new surge of magma moving eastward laterally through the sill where immiscible sulfide liquid from the previous magma was deposited (Fig. 14a, b), similar to the process proposed for the Voisey's Bay deposit by Li and Naldrett (1999). The two spatially closely associated IPGE-depleted ore zones (I-a, II-W-a) formed by a more fractionated, sulfide-charged magma (Fig. 14c). After solidification, the deposit was cut by a series of faults into different segments (Fig. 2a). The two IPGE-depleted ore zones were once a single ore zone before faulting.

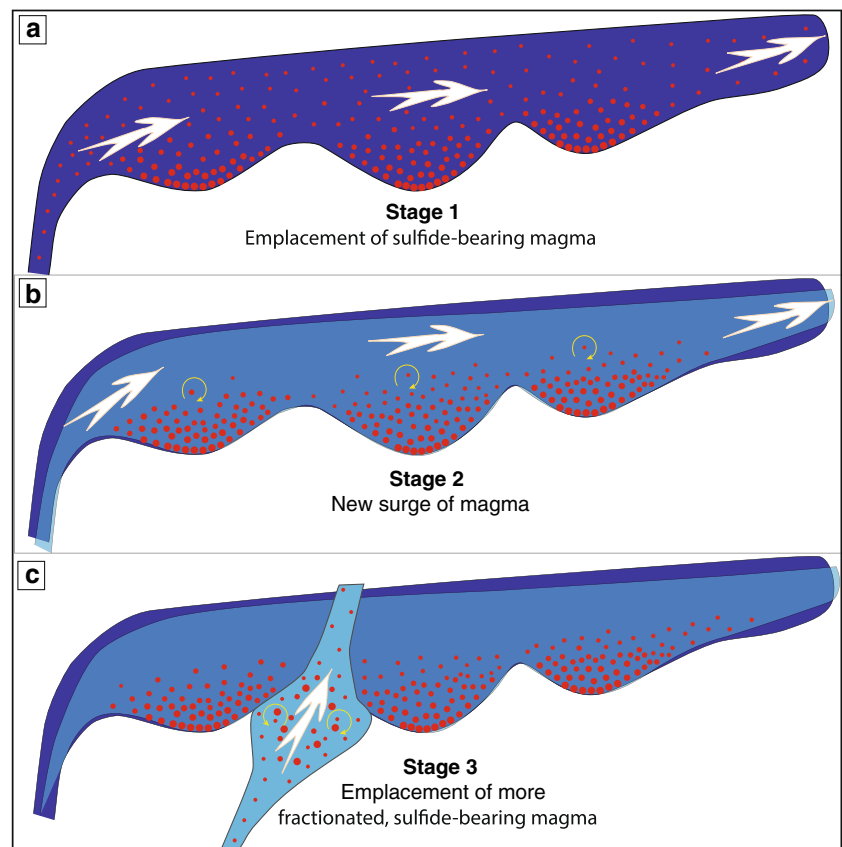
The variations of PGE tenors in the Jinchuan deposit can also be explained by the emplacement of multiple sulfide liquids (Tang 1993) or multiple pulses of

sulfide-charged magma with different R-factors at depth (Chai and Naldrett 1992b). However, such models require a dyke-like geometry for each of the ore zones that have different PGE tenors. Detailed mapping shows that the original geometry of the Jinchuan intrusion is more likely a subhorizontal sheet for most parts of the intrusion (Lehmann et al. 2007).

We propose that the three different pulses of magma were all charged with olivine crystals upon arrival at Jinchuan, because the bulk of the Jinchuan intrusion is composed of olivine cumulates (Chai and Naldrett 1992a; de Waal et al. 2004; Li et al. 2004). In addition, we suggest that minor immiscible sulfide droplets were present in the first and last pulses of magma before they arrived at Jinchuan, because the sulfide/olivine ratios in the related ore zones far exceed the cotectic sulfide/ratios expected from sulfide-saturated magma that does not contain excessive sulfide (Li and Ripley 2011).

As discussed above, the Jinchuan sulfide ores likely formed by the second stage of sulfide segregation. This interpretation, together with the observed trend of increasing metal tenors from east to west laterally, implies that the area beneath the western part of segment I is the best exploration target for high metal tenor sulfide ores formed by previous sulfide segregation in the Jinchuan magma plumbing system.

Fig. 14 a–c A dynamic ore genesis model for the Jinchuan deposit. See text for explanations



Conclusions

The most important observations and conclusions from this study are summarized below:

1. The Jinchuan deposit can be divided into two separate magmatic systems: one is represented by two spatially related, relatively IPGE-depleted sulfide ore zones (I-a, II-W-a) in the western part of the deposit, and the other is represented by the rest of the deposit which formed earlier.
2. The bulk sulfide ores of the early magmatic system are characterized by a lateral, eastward decrease in PGE tenors, which can be explained by upgrading of sulfide liquid in a subhorizontal conduit by a new surge of magma moving through the conduit from west to east.
3. IPGE depletion in the bulk sulfide ores of the late magmatic system is coupled by depletion in Ni tenor and more fractionated Cr-spinel and olivine compositions than those of the early system, indicating a more fractionated parental magma for the late magmatic system.
4. Modeling results indicate that fractional crystallization did not play a major role in triggering sulfide saturation in the Jinchuan magma. Hf-Sr-Nd and multiple S isotopes reveal that crustal contamination was important in the formation of the Jinchuan deposit.

Acknowledgments This study was financially supported by the National Science Foundation of China (41372101, 41072058), China Geological Survey (12120114044401, 1212011085061), Special Funds for Public Welfare Project (200911007), and Chang'an University (2013G2273009, 310826151138). We thank Yaling Gao, Hao Lu, and Yushan Wang for assistance in field work and Anne Ayre and Benjamin Underwood for assistance in multiple S isotope analysis. This paper was written during Jun Duan's visit to Indiana University which was financially supported by the China Scholarship Council (201306560011). Constructive reviews from Nicholas Arndt and Marco Fiorentini are greatly appreciated.

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